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ReHyAt: Recurrent Hybrid Attention for Video Diffusion Transformers

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Abstract

Recent advances in video diffusion models have shifted towards transformer-based architectures, achieving state-of-the-art video generation but at the cost of quadratic attention complexity, which severely limits scalability for longer sequences. We introduce ReHyAt, a Recurrent Hybrid Attention mechanism that combines the fidelity of softmax attention with the efficiency of linear attention, enabling chunk-wise recurrent reformulation and constant memory usage. Unlike the concurrent linear-only SANA Video, ReHyAt's hybrid design allows efficient distillation from existing softmax-based models, reducing the training cost by two orders of magnitude to ∼160 GPU hours, while being competitive in the quality. Our light-weight distillation and finetuning pipeline provides a recipe that can be applied to future state-of-the-art bidirectional softmax-based models. Experiments on VBench and VBench-2.0, as well as a human preference study, demonstrate that ReHyAt achieves state-of-the-art video quality while reducing attention cost from quadratic to linear, unlocking practical scalability for long-duration and on-device video generation.

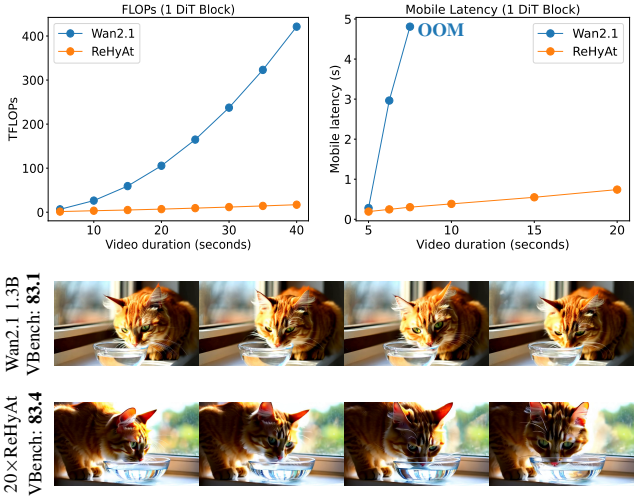


Figure 1. A comparison of our proposed Recurrent Hybrid Attention model with Wan2.1 bidirectional full softmax attention. Top: Compute complexity increase with video duration growth (left: FLOPs, right: phone latency). Bottom: comparing our hybrid model (20× ReHyAt blocks) with original Wan2.1 1.3B, qualitatively and quantitatively. Prompt: "A cat drinking water."

1. Introduction

The ambition in generative video is shifting from producing short, visually striking clips to creating sustained, coherent sequences with rich dynamics and consistent subject identity. Diffusion-based models have become the method of choice for this goal due to their stability and controllability; however, the choice of backbone is decisive for scaling. While early video diffusion systems adapted U-Net architectures from images, they exhibited limited capacity to model long temporal structure and struggled to scale effectively to higher resolutions and durations. This has motivated a transition to Diffusion Transformers (DiTs) [30], which process video as a sequence of spatiotemporal patches and furnish global context from the first layer. The resulting architectural shift underlies recent state-of-the-art systems (e.g., *Wan2.1* [35], *CogVideoX* [44], *HunyuanVideo* [22], *PyramidalFlow* [16], *Open-Sora Plan* [26]), and has been documented by recent surveys as the prevailing trend in video generation [28, 39].

This progress comes with a nontrivial systems cost: the self-attention term scales quadratically with sequence length, $\mathcal{O}(N^2 d)$ in time and $\mathcal{O}(N^2)$ in memory, where N is the number of tokens and d the hidden dimension [33, 34]. In video, N is the product of temporal length and spatial patch count, so even moderate resolutions and durations yield token counts in the tens of thousands. In practice, the attention subroutine consumes the majority of compute in DiT blocks, and memory pressure grows rapidly with longer contexts. Kernel- and IO-aware implementations such as FlashAttention [7] reduce constants but do not alter the N^2 dependence, leaving training and inference constrained when targeting higher resolutions, extended durations, or multi-shot compositions. As a direct consequence,

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producing videos beyond roughly 10 seconds remains difficult within typical GPU memory and latency budgets, while edge devices such as mobile phones even struggle to generate more than a few seconds of videos.

Linear attention [20] offers a compelling alternative to full softmax attention by reducing complexity from quadratic to linear and enabling constant memory when reformulated as an RNN. This property makes it particularly attractive for generating arbitrarily long videos, where memory growth is a critical bottleneck. Beyond efficiency, the recurrent formulation of linear attention allows chunk-wise processing, which aligns naturally with sequential video generation. These advantages have motivated recent efforts to explore linear and hybrid attention mechanisms in video diffusion models [4, 11].

However, linear attention introduces a significant trade-off: its kernel-based similarity function lacks the expressiveness of the exponential kernel used in softmax attention. This gap manifests in reduced activation diversity and weaker modeling of fine-grained dependencies [47], often requiring extensive retraining to achieve acceptable quality [4, 11]. Hybrid approaches that combine linear and softmax attention have emerged as a potential solution [11], but existing designs remain quadratic in complexity and cannot be reformulated as RNNs, leaving the scalability challenge unresolved. In other words, while these methods improve quality over purely linear attention, they fail to deliver the memory and compute benefits necessary for long-duration video generation.

Meanwhile, the most powerful video diffusion models today are trained with bidirectional full softmax attention using massive compute and data resources. Re-training such models with alternative attention mechanisms from scratch is prohibitively expensive and impractical for most research and production settings. This observation motivates a different strategy: rather than building efficient models from the ground up, can we distill these high-quality, compute-heavy models into a recurrent form that preserves fidelity while dramatically reducing resource requirements? Achieving this would unlock practical scalability for video diffusion, not neglecting the substantial progress made by state-of-the-art architectures.

In this paper, we address this challenge by introducing **ReHyAt**, a recurrent hybrid attention mechanism tailored for video diffusion. Our key insight is that preserving softmax attention for a small subset of tokens—those most critical for modeling local dependencies—while applying linear attention globally enables modeling long-range and high fidelity local dependencies while ensuring linear efficiency. We propose a temporally chunked hybrid attention design with overlapping chunks to maintain temporal coherence, and show that this formulation can be reformulated into a chunk-wise RNN with constant memory complexity. Furthermore, we leverage a two-stage training pipeline—attention distillation from a bidirectional softmax teacher followed by lightweight fine-tuning—that achieves SOTA results within fewer than 200 GPU-hours. We validate our approach by transforming *Wan2.1* into its recurrent hybrid counterpart and evaluate on VBench [14], VBench2.0 [51], and a human preference study, demonstrating that ReHyAt delivers near state-of-the-art quality with dramatically reduced compute. Fig 1 demonstrates some of the aspects discussed above.

Our main contributions are as follows:

- We propose *ReHyAt*, a novel temporally chunked hybrid attention mechanism that combines local softmax attention with global linear attention. This design preserves high-fidelity modeling of critical dependencies within and across adjacent frames while reducing overall complexity to linear time.
- We derive a chunk-wise *recurrent* reformulation of ReHyAt, computationally enabling generation of arbitrarily long videos with constant memory usage and efficient inference.
- Through extensive empirical evaluations and ablation studies, we show that a state-of-the-art bidirectional Softmax attention video diffusion model can be transformed into a chunk-wise recurrent model, only within a few hundred GPU-hours, with negligible impact on the quality.

2. Related Work

Efficient Attention. Several approaches aim to reduce the quadratic complexity of self-attention across domains: for vision tasks (e.g., EfficientViT [2], PADRe [23], Performer [5], Linformer [38]), image generation (e.g., SANA [25], LinGen [36], Grafting [3]), and language modeling [29, 37, 43, 46, 48]. These works show the feasibility of sub-quadratic attention but often require heavy retraining or training from scratch (e.g., SANA [25]). In contrast, we focus on lightweight distillation and fine-tuning of pre-trained softmax-based models into an efficient hybrid attention design tailored for video diffusion under modest compute budgets. Linear recurrent models such as SSM and RWKV [9, 10, 37, 45, 53] have emerged as alternatives to self-attention for long sequences. However, architectural differences from transformers make distilling DiT weights into these models costly. Our approach preserves the original block structure, enabling effective distillation with minimal training. Finally, as noted in Katharopoulos et al. [20], causal linear attention can be reformulated as an RNN during inference—a property we leverage for efficient long video generation.

Video Diffusion Models. Recent large-scale systems such as CogVideoX [44], Open-Sora Plan [26], PyramidalFlow [16], LTX-video [12], and Wan2.1 [35] have significantly advanced video generation quality and scalability,

but at substantial compute and memory cost. Mobile/PC-oriented designs like Mobile Video Diffusion [42], MoViE [18], SnapGen-V [41], AMD-HummingBird [15], On-device Sora [21], MobileVDiT [40], and Neo-Dragon [19] aim for lightweight deployment, yet most remain non-DiT-based or still rely on full quadratic attention, limiting scalability for long-duration videos.

Video Diffusion Models with Efficient Attention. Prior work has explored accelerating video generation through token merging [1, 8, 17], token downsampling [6, 31], attention tiling [8, 50], and sparsity [24, 49]. Tiling and sparsity-based approaches, in particular, gain efficiency by discarding attention for most tokens. In contrast, our hybrid attention design attends to the full token set, combining linear attention for long-range dependencies with softmax attention for local, high-fidelity interactions. M4V [13] accelerates video DiTs by distilling them into Mamba blocks. Despite our simpler block structure and lightweight training, we outperform M4V in both quality and efficiency.

Recently, Attention Surgery [11] proposed a temporally uniform hybrid attention method with reasonable quality but retained quadratic complexity. Our approach introduces a temporally non-uniform hybrid arrangement, enabling uneven treatment of token dependencies and a better inductive bias for video generation. It achieves linear complexity and can be reformulated as a memory-efficient RNN, supporting on-device execution and scalable long video generation.

Finally, concurrent to our work, SANA-Video [4] introduced a video diffusion model incorporating linear attention. In contrast, our method offers a hybrid approach combining the computational efficiency of linear attention for long-range dependencies with the accuracy of softmax attention for modeling highly co-dependent adjacent tokens. Furthermore, unlike SANA-Video, our method sets up a distillation process from a SOTA bidirectional full softmax attention model, making training extremely efficient: we obtain our model in ∼160 GPU-hours—*two orders of magnitude more efficient than SANA-Video*. This work therefore provides a low-cost recipe to transform costly Softmax attention SOTA models into efficient RNNs, laying the groundwork for long video generation and on-device execution.

3. Methods: ReHyAt

3.1. Preliminaries: Linear Attention

Let $x \in \mathbb{R}^{N\times D}$ denote a sequence of N tokens, each represented by a D-dimensional feature vector. At the l-th transformer layer, the block is formulated as:

$$T_l(x) = f_l\big(A_l(x) + x\big), \tag{1}$$

where $f_l(\cdot)$ applies a token-wise transformation, typically a lightweight feed-forward network, and $A_l(\cdot)$ represents the self-attention operator—the component responsible for cross-token interaction. The standard attention mechanism is given by:

$$A_l(x) = y = \text{softmax}\left(\frac{qk^\top}{\sqrt{D}}\right) v, \tag{2}$$

where queries, keys, and values are computed as linear projections:

$$q = xw_q,\ k = xw_k,\ v = xw_v,$$

with learnable weights $w_q, w_k, w_v \in \mathbb{R}^{D\times D}$.

The softmax attention for token i can be expressed as:

$$y_i = \frac{\sum_{j=1}^{N} \text{sim}(q_i, k_j)\, v_j}{\sum_{j=1}^{N} \text{sim}(q_i, k_j)}. \tag{3}$$

Applying the kernel trick, the similarity function can be generalized from $\text{sim}(q_i, k_j) = e^{q_i k_j^\top}$ (recovering the original softmax) to $\text{sim}(q_i, k_j) = \phi(q_i)\phi(k_j)^\top$, yielding:

$$y_i = \frac{\phi(q_i) \sum_{j=1}^{N} \phi(k_j)\, v_j^\top}{\phi(q_i) \sum_{j=1}^{N} \phi(k_j)}. \tag{4}$$

Crucially, the terms $\sum_{j=1}^{N} \phi(k_j)\, v_j^\top$ and $\sum_{j=1}^{N} \phi(k_j)$ do not depend on i, enabling precomputation and caching for linear-time complexity. The mapping $\phi(\cdot)$ must be non-negative; the original work by Katharopoulos et al. [20] proposes $\phi(x) = 1 + \text{elu}(x)$. However, this substitution introduces a notable gap in expressiveness compared to the exponential kernel, often requiring substantial retraining or resulting in degraded performance relative to softmax attention.

3.2. Hybrid Attention Formulation

Before introducing the formal expression, we note that the hybrid attention mechanism combines contributions from both softmax attention (for local, high-fidelity dependencies) and linear attention (for global, efficient interactions), and normalizes them jointly.

For the latent $x \in \mathbb{R}^{N\times D}$, assume the N tokens are flattened from a latent tensor of shape (T, H, W, D), where $N = THW$. To overcome the limitations of purely linear attention in video diffusion models, we incorporate a *hybrid attention* mechanism that combines softmax-based and kernelized linear attention formulations. Now consider a chunk of T_c temporal slices from the latent, represented as $X_t \in \mathbb{R}^{N'\times D}$, where $N' = T_c HW$. Here we have introduced the chunk-indexed reshaped notation $X \in \mathbb{R}^{T'\times N'\times D}$, with $T' = N/N'$ representing the number of chunks, to avoid confusion with single token indexing e.g. x_i. Following the same notation, we have $Q_t \in \mathbb{R}^{N'\times D}$, and $\phi_q(Q_t) \in \mathbb{R}^{N'\times D'}$. Then for the hybrid attention of tokens in chunk t, we partition the total tokens

$\mathcal{T} = \{1, 2..., N\}$ to attend to, into softmax attention tokens $\mathcal{T}_t^{\text{S}}$ and linear attention tokens $\mathcal{T}_t^{\text{L}}$. More specifically, the hybrid attention output for token chunk t, $\hat{y}_t \in \mathbb{R}^{N' \times D}$ constitutes of softmax attention and its normalizer $a_t^{\text{S}} \in \mathbb{R}^{N' \times D}$ and $n_t^{\text{S}} \in \mathbb{R}^{N' \times 1}$ as well as linear attention and its normalization term $a_t^{\text{L}} \in \mathbb{R}^{N' \times D}$, $n_t^{\text{L}} \in \mathbb{R}^{N' \times 1}$, formulated as below:

$$\hat{y}_t = \frac{a_t^{\text{S}} + a_t^{\text{L}}}{n_t^{\text{S}} + n_t^{\text{L}}}, \tag{5}$$

$$a_t^{\text{S}} = \sum_{j \in \mathcal{T}_t^{\text{S}}} \exp(Q_t k_j^\top / \sqrt{D} - c_t) v_j, \tag{6}$$

$$a_t^{\text{L}} = \phi_q(Q_t) \Big(\sum_{j \in \mathcal{T}_t^{\text{L}}} \phi_k(k_j)\, v_j^\top \Big), \tag{7}$$

$$n_t^{\text{S}} = \sum_{j \in \mathcal{T}_t^{\text{S}}} \exp(Q_t k_j^\top / \sqrt{D} - c_t), \tag{8}$$

$$n_t^{\text{L}} = \phi_q(Q_t) \Big(\sum_{j \in \mathcal{T}_t^{\text{L}}} \phi_k(k_j) \Big), \tag{9}$$

where c_t is a stabilizing constant (typically the maximum exponent), and $\phi_q(\cdot)$ and $\phi_k(\cdot)$ denote the kernel feature maps for the linear component for queries and keys.

Here, we propose the following specification for the partitioning of the tokens sets:

$$\mathcal{T}_t^{\text{S}} = \{j \mid tN' \leq j < (t+1)N'\}, \tag{10}$$

$$\mathcal{T}_t^{\text{L}} = \mathcal{T} - \mathcal{T}_t^{\text{S}} \tag{11}$$

See the top graph in Fig 2. This means that the computation of attention is effectively broken into temporal chunks of T_c slices, where the tokens within each slice more accurately attend to each other with the Softmax attention, and with linear attention to all the other tokens.

Overlapping Chunks. We observe that the non-overlapping chunking mechanism defined above, together with the lower fidelity dependency modeling of linear attention, can result into episodic incoherence in motion or appearance between the frames transitioning from one latent chunk to next. To mitigate this, we propose to arrange overlapping chunks for softmax attention, enabling a more accurate softmax-attention-based message passing between the chunks. More specifically, for generating attention output for a chunk given chunk of T_c slices (i.e. applying this slicing to queries), the keys and values representing the tokens to attend to, are sliced by $T_c + T_o$ temporal slices instead, where T_o represents the overlap size. To arrange this, one needs to reformulate $\mathcal{T}_t^S$ and $\mathcal{T}_t^L$ as:

$$\begin{aligned} \mathcal{T}_t^{\text{S}} &= \{j \mid \max(tN' - T_o HW, 0) \leq j < (t+1)N'\} \\ \mathcal{T}_t^{\text{L}} &= \mathcal{T} - \mathcal{T}_t^{\text{S}} \end{aligned} \tag{12}$$

The bottom subgraph in Fig 2 illustrates this.

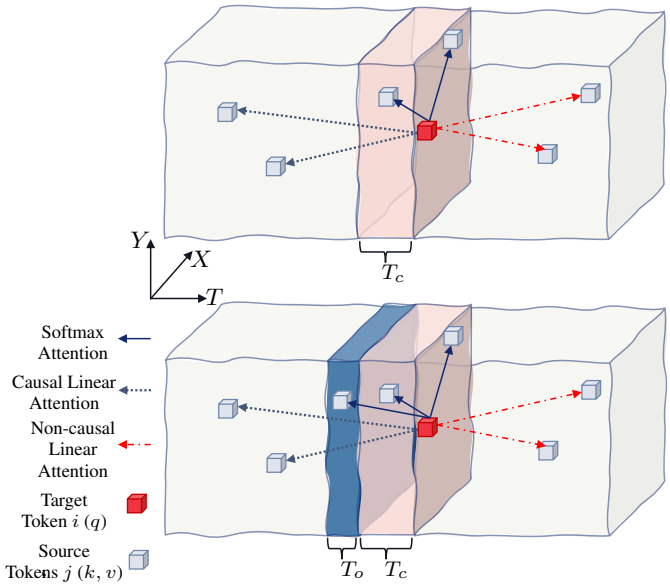


Figure 2. Overview of the temporally chunked hybrid attention arrangement without (top) and with chunk overlap (bottom).

Characterization of ϕ. Similar to [11], to enhance the expressiveness of linear attention, we define distinct learnable feature maps $\phi_q, \phi_k : \mathbb{R}^D \to \mathbb{R}^{D'}$. Each map first applies a lightweight per-head embedding network (implemented as grouped 1×1 convs with non-linear activations) to produce an intermediate representation, which is then split into P equal parts. Each part is raised to a different polynomial degree 1 to P, and concatenated along the feature dimension. Formally, for an input $x \in \mathbb{R}^D$, we define:

$$\phi(x) = [\ (\psi_1(x))^1, (\psi_2(x))^2, \ldots, (\psi_P(x))^P\ ]^\top \in \mathbb{R}^{D'},$$

where $\psi_i(\cdot)$ denotes the i-th learnable embedding slice produced by the shared embedding network. This polynomial expansion allows $\phi_q(q_i)\phi_k(k_j)^\top$ to approximate the large dynamic range of the exponential kernel $e^{q_i k_j^\top}$ more accurately than fixed ELU-based mappings.

3.3. Recurrent HyAt

Linear attention, once causal has the advantage that can be reformulated to RNNs. In this section we show how our hybrid arrangement, unlike [11], can be reformulated as an RNN. For this to be feasible, we first need to make it causal. To achieve this, it is sufficient to reformulate $\mathcal{T}_t^{\text{L}}$ as follows:

$$\begin{aligned} \mathcal{T}_t^{\text{L}} &= \{j \mid j < \max(tN' - T_o HW, 0)\} \\ \mathcal{T}_t^{\text{S}} &= \{j \mid \max(tN' - T_o HW, 0) \leq j < (t+1)N'\} \end{aligned} \tag{13}$$

In Fig. 2, this is equivalent to the bottom graph where the specified non-causal linear attention is removed. Now thanks to the temporal decoupling of $\mathcal{T}_t^{\text{S}}$ and $\mathcal{T}_t^{\text{L}}$, we can

define a chunk-wise RNN, where the model generates the latents chunk-by-chunk for T_c temporal slices at a time. Let $s_t \in \mathbb{R}^{D' \times D}$ and $z_t \in \mathbb{R}^{D' \times 1}$ represent the state variables for the linear attention and its normalizer in t-th chunk. Then we have:

$$s_0 = 0 \tag{14}$$

$$z_0 = 0 \tag{15}$$

$$y_t = \frac{a_t^{\mathrm{S}} + \phi_q(Q_t)\, s_t}{n_t^{\mathrm{S}} + \phi_q(Q_t)\, z_t} \tag{16}$$

$$s_{t+1} = s_t + \sum_{j \in \mathcal{T}_t^{\mathrm{S}}} \phi_k(k_j)\, v_j^\top \tag{17}$$

$$z_{t+1} = z_t + \sum_{j \in \mathcal{T}_t^{\mathrm{S}}} \phi_k(k_j) \tag{18}$$

Three points to note: (1) Softmax attention within each chunk need not be causal because sampling proceeds chunk-by-chunk, i.e. our method generates the latents for a full chunk at once. (2) The model training doesn't have to be done in the RNN form as introduced above. One can train the model in the causal non-recurrent form and then rearrange the trained model to RNN at the sampling time. (3) The computational complexity remains $\mathcal{O}(N)$ with the length of the generated video, while the memory complexity remains constant irrespective of the video duration.

3.4. Two-stage Training

Given the enormous compute/data requirements for obtaining SOTA video diffusion models, our proposed method is instead centered around efficiently distilling existing bidirectional full softmax attention, e.g. Wan2.1, into our proposed RNN formulation. To achieve this, we propose a two-stage process: attention distillation and lightweight finetuning that we expand in the following. Thanks to this specific method design, we obtain a recurrent video diffusion model with competitive quality, within less than 200 GPU-hours.

3.4.1. Attention Distillation

We first distill a bidirectional full softmax teacher model into a causal hybrid attention student model. During this stage, each block is trained independently and the only learnable parameters are ϕ_q and ϕ_k per block, so as to let ϕ parameters to enable linear attention to approximate the corresponding softmax dependencies. This distillation setup doesn't require any prompt/video pairs for the training; the student model is trained to match the teacher activations for different prompts, noise samples and denoising iterations. The following equation formalizes this:

$$\boldsymbol{\phi}_l = \boldsymbol{\phi}_l - \eta \nabla_{\boldsymbol{\phi}_l} \left(\mathbb{E}_{\substack{\epsilon \in \mathcal{N} \\ p \in \mathcal{P} \\ i \in \mathcal{S}}} \left| y^{(l,\epsilon,p,i)} - \hat{y}^{(l,\epsilon,p,i)} \right| \right), \tag{19}$$

where $\boldsymbol{\phi}_l$ is (ϕ_q, ϕ_k) for the l-th block, $\mathcal{N}$ the noise sampling distribution, $\mathcal{P}$ the distribution of textual prompts, $\mathcal{S}$ the set of denoising steps, $y^{(l,\epsilon,p,i)}$ the output of the bidirectional softmax teacher on block l, for prompt p, sampling noise ϵ, and denoising step i, and $\hat{y}^{(l,\epsilon,p,i)}$ the same trajectory point for the ReHyAt student model.

3.4.2. Lightweight Fine-tuning

After the pretraining distillation stage making the block attentions recurrent hybrid, we have obtained the ϕ_qs and ϕ_ks per block. However, while the pretraining distillation helps preserve the general structure of the scenes, the details will be far from perfect, specifically on the transition smoothness between chunks, as the blocks are pretrained in isolation. Now fine-tuning the whole DiT model on a modest set of prompt/video pairs, for a small number of iterations (e.g. 1k) recovers the lost generation quality. This is done by optimizing the normal flow-matching objective [27].

4. Experimental Setup

4.1. Evaluation of generation quality

We evaluate ReHyAt by distilling and fine-tuning Wan2.1 1.3B model [35], a widely used efficient SOTA model. For SOTA comparisons, we generate videos at the original Wan resolution and length ($81 \times 480 \times 832$) using the full set of extended prompts from VBench [14] and VBench-2.0 [51].

In addition to quantitative evaluation, we conduct a blinded human preference study to assess visual qualities and prompt alignment. We randomly select 50 prompts from VBench and present participants with paired videos, asking them to choose their preferred video or indicate no significant difference. The order of paired videos randomly change per prompt to avoid any potential biases. In total, we collect 500 paired comparisons.

To enable large-scale ablation studies, we train and evaluate our model variants at a lower spatial resolution of 320×480 per frame. For all evaluations, we use the model snapshot at the 1000th fine-tuning iteration.

4.2. Assessment of compute complexity

FLOPs Analysis. We analyze the number of floating point operations in the proposed ReHyAt method and compare it against flash attention, and other alternatives on the original 5-second Wan video generation setup, as well as analyzing the DiT blocks' compute growth as we increase the length of generated videos. For this, we use the DeepSpeed library to measure the complexities.

On-mobile Measurements. A valuable advantage of the proposed recurrent hybrid attention method is that it computationally enables the generation of longer videos on edge devices such as mobile phones, thanks to lower compute burden, and most importantly, due to significant reduction in peak memory consumption. We port the transformed original WAN model with flash attention blocks as well as

Models with 2B–5B parameters	Total↑	Quality↑	Semantic↑
Open-Sora Plan V1.3 [26]	77.23	80.14	65.62
CogVideoX 5B [44]	81.91	83.05	77.33
CogVideoX1.5 5B [44]	82.01	82.72	79.17
Models up to 2B parameters			
Open-Sora V1.2 [52]	79.76	81.35	73.39
LTX-Video [12]	80.00	82.30	70.79
SnapGenV [41]	81.14	83.47	71.84
Hummingbird 16frame [15]	81.35	83.73	71.84
Mobile Video DiT - Mobile [40]	81.45	83.12	74.76
Mobile Video DiT - Server [40]	83.09	84.65	76.86
CogVideoX 2B [44]	81.55	82.48	77.81
PyramidalFlow [16]	81.72	84.74	69.62
Neodragon [19]	81.61	83.68	73.36
Wan2.1 1.3B [35]	83.31	85.23	75.65
Wan2.1 1.3B* [35]	83.10	85.10	75.12
Linear/Hybrid Models			
Efficient VDiT [8]	76.14	-	-
M4V [13]	81.91	83.36	76.10
STA [50]	83.00	**85.37**	73.52
VSA [49]	82.77	83.60	79.47
SANA-Video [4]	83.71	84.35	**81.35**
Attention Surgery (15×R2) [11]	83.21	85.19	75.25
Wan2.1 1.3B* + ReHyAt (15×T_c=3,T_o=1)	**83.79**	84.57	80.70

Table 1. Comparisons with SOTA efficient video diffusion models. 'Wan2.1*' is our best reproduction using our evaluation pipeline.

Model	VBench-2.0					
	Total↑	Hum.Fid.↑	Creativity↑	Control.↑	Com.sense↑	Physics↑
Wan2.1 1.3B	56.0	80.7	48.7	**34.0**	63.4	**53.8**
CogVideoX-1.5 5B	53.4	72.1	43.7	29.6	63.2	48.2
Attn. Surgery 15×R2	55.1	78.9	47.5	33.4	63.1	52.8
ReHyAt 15×T_c=3	56.1	**81.9**	55.1	30.8	62.7	50.0
ReHyAt 15×T_c=5	**56.3**	79.8	**55.7**	31.9	**64.2**	49.7

Table 2. Quantitative comparison on VBench-2.0 benchmark

Prompt Dimension	Human Preference %		
	Ours	No preference	Wan2.1
Color	43.3	46.7	10.0
Human Action	21.7	41.7	36.7
Object Class	25.0	45.0	30.0
Overall Consistency	27.1	47.1	25.9
Scene	40.0	60.0	0.0
Spatial Relationship	20.0	70.0	10.0
Subject Consistency	21.7	28.3	50.0
Temporal Flickering	24.0	54.0	22.0
Temporal Style	43.3	30.0	26.7
Total	27.6	43.5	29.0

Table 3. Results of the method-blinded human visual preference study over 500 paired video comparisons. Rows correspond to subsets filtered by different VBench prompt dimensions receiving 10+ prompt comparisons.

the transformed ReHyAt modules to Qualcomm AI Runtime (QNN) and profile run-time metrics such as latency, memory read, and memory write on a Snapdragon8-Gen4 SoC. For the on-device measurements we report the metrics on 320×480 frame resolution with the original 5 seconds WAN video length, as well as longer video durations.

4.3. Training specification

Datasets. For fine-tuning low-resolution models, we use a 350K subset of the video dataset from Open-Sora Plan [26]. For high-resolution fine-tuning, we use 22K synthetic video samples generated by Wan2.1 14B, with prompts drawn from the same source as used for the low-resolution dataset.
Model Hyperparameters. We experiment with converting different numbers of transformer blocks to recurrent hybrid attention: 15, 20, and 25 out of the 30 blocks in Wan2.1 1.3B. For the hybrid blocks, we explore hybridization with various chunk sizes ($T_c \in \{1, 2, 3, 5, 7\}$) as well as different options for overlap size ($T_o \in \{0, 1, 2, 3\}$). Empirical analysis of the impact of ϕ_k and ϕ_q transformation complexity on generation quality shows that a lightweight 2-layer MLP with degree-2 polynomial features is sufficient. This configuration adds approximately 2.4M parameters per converted block. Additional details are provided in the appendix.

5. Results

5.1. Generation Quality

VBench SOTA. Table 1 compares ReHyAt model distilled from Wan2.1 1.3B against the state-of-the-art efficient video diffusion models up to 5B parameters. We observe that our method performs very competitively, while forming a chunk-wise RNN that enables running it on mobile. Note that the compute burden to obtain our model is ∼160 H100 GPU hours, i.e. less than 1% of SANA-Video (12 days of 64 H100) and less than 0.01% of MovieGen [32].
VBench-2.0. Table 2 presents the evaluation and comparison of SOTA methods on VBench-2.0 benchmark. While we observe a small drop, ReHyAt still remain competitive to larger models such as CogVideoX1.5 5B.
Human Preference Evaluation. Table 3 shows the results for the human visual preference study comparing our 15×T_c=3 model against the original Wan2.1 1.3B, from a total of 500 paired video comparisons. As can be observed, there is no significant difference between our recurrent hybrid model and the original Wan2.1 in human preference.

Figure 3 shows two qualitative samples and how ReHyAt compares to Wan2.1 1.3B. More extensive set of qualitative samples are provided in the supplementary materials.

5.2. Sampling Compute Burden

In Fig. 4 we measure how chunk size T_c and chunk overlap size T_o impact the number of floating point operations, also how it compares to flash attention and uniform hybrid attention [11] with various rates $R \in \{2, 4, 8\}$, as measured on 5s videos at 480×832 resolution, corresponding to a latent size 21×30×52. As can be observed, ReHyAt offers up to 4× operation saving as compared to flash attention used by Wan2.1. On the other hand, our T_c=3, T_o=1 model variant

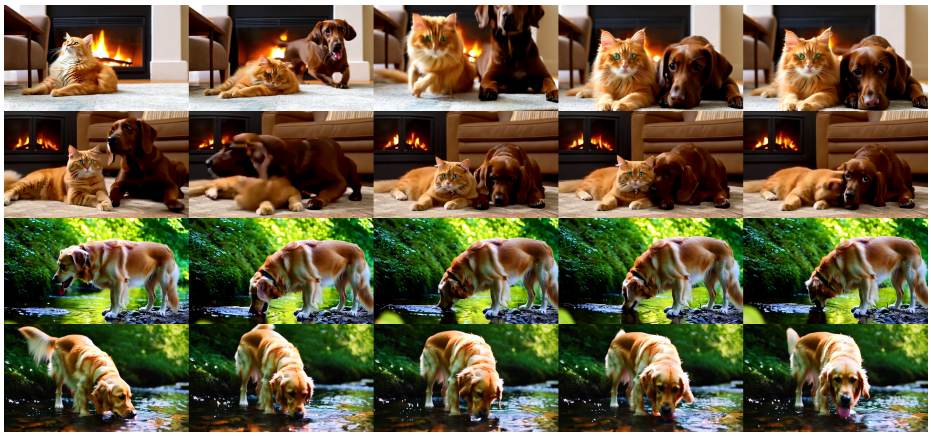


Figure 3. Qualitative comparison of Wan2.1 1.3B (Top) to ReHyAt 15×T_c=3 (bottom) for two sample VBench prompts, "A cat and a dog." and "A dog drinking water."

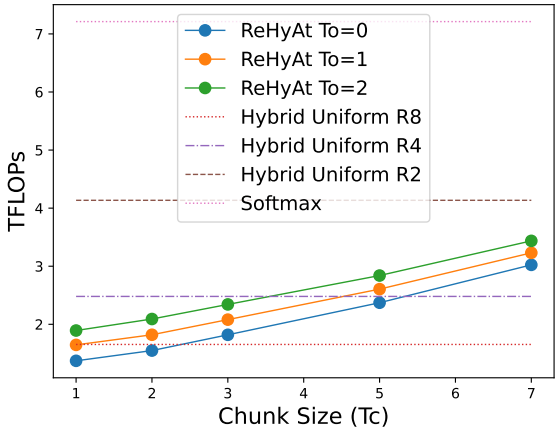


Figure 4. Comparison of attention compute (FLOPs) on 21×30×52 latent size (5 seconds)

Attention Block	Number of frames (320×480) resolution				
	81	101	121	141	161
Softmax Flash Attention	281	2964	4809	OOM	OOM
HedgeHog Linear Attention	360	455	469	542	OOM
Uniform Hybrid - R8	464	625	818	1215	OOM
ReHyAt - T_c=3 (ours)	**192**	**247**	**302**	**329**	**384**

Table 4. On mobile (Snapdragon8-Gen4) latency (ms) vs. number of frames at 320×480 resolution

remains ∼2× more efficient as compared to the better quality preserving $R = 2$ uniform hybrid attention variation.

Fig. 1 top demonstrates how the compute burden grows with increased video duration, comparing the scaling behavior for the Wan2.1 1.3B (flash attention) versus our proposed method (ReHyAt). Here we see that compared to flash attention, our recurrent hybrid attention has a significantly better scaling behavior.

Table 4 presents the on-mobile, DiT block latencies in miliseconds for various types of attention mechanism, flash attention, HedgeHog linear attention with learnable ϕ, uniform hybrid attention R=7 and our ReHyAt hybrid method with T_o=3, for various video durations from 5s (81 frames) to 10s (161 frames). As can be observed, our recurrent hybrid method is the only one that can easily extend to more than 10s without out-of-memory errors. Within the feasible extent for flash attention (e.g. on 121 frames), our method is ∼16× faster than flash attention used in Wan2.1 1.3B.

Table 5 shows the memory read/write load that correlates with power consumption and latency. As we observe, due to its more memory-efficient design, our recurrent hybrid attention model is significantly more memory-efficient, e.g. ∼11×more efficient in total memory read/write than flash attention at 121 frames (∼7.5s duration). Please note that while the total memory/read write is expected to grow linearly with video duration for ReHyAt, the peak-memory usage remains constant.

5.3. Ablations Studies

Number of ReHyAt Blocks and Chunk-size T_c. Fig. 5 shows scatter plots comparing the computational cost of different variations of ReHyAt, with various number of converted blocks and chunk-sizes T_c as well as the original Wan2.1 1.3B model, at both 320×480 and 480×832 resolutions. Table 6 shows the Vbench full set evaluation for various T_c's. As expected, increasing T_c generally improves model quality; however, the increase from 1 to 2 yields a more significant improvement compared to further increases to 3 and 4. This is perhaps due to the first extension of the softmax from spatial to spatiotemporal.

Overlap size T_o. Table 7 demonstrates how different chunk overlap size T_o values (ranging from 0 to 3) impacts the generation quality. As anticipated, enabling overlap (i.e., going from T_o = 0 to T_o = 1) results in a notable jump in

Attention Block	Number of frames - Memory Read/Write (GB)									
	81		101		121		141		161	
	W	R	W	R	W	R	W	R	W	R
Softmax Flash Attention	5.1	6.0	12.9	16.4	22.7	53.6	OOM	OOM	OOM	OOM
HedgeHog Linear Attention	5.7	8.1	7.0	10.1	6.9	11.3	8.0	13.2	OOM	OOM
Uniform Hybrid - R8	6.3	10.1	5.2	10.9	6.4	13.2	7.8	35.2	OOM	OOM
ReHyAt - T_c=3 (ours)	**1.7**	**2.8**	**2.2**	**3.6**	**2.7**	**4.4**	**3.0**	**4.8**	**3.5**	**5.6**

Table 5. Comparison of total memory read/write for Wan2.1 DiT Blocks with various attention mechanisms on Snapdragon8-Gen4

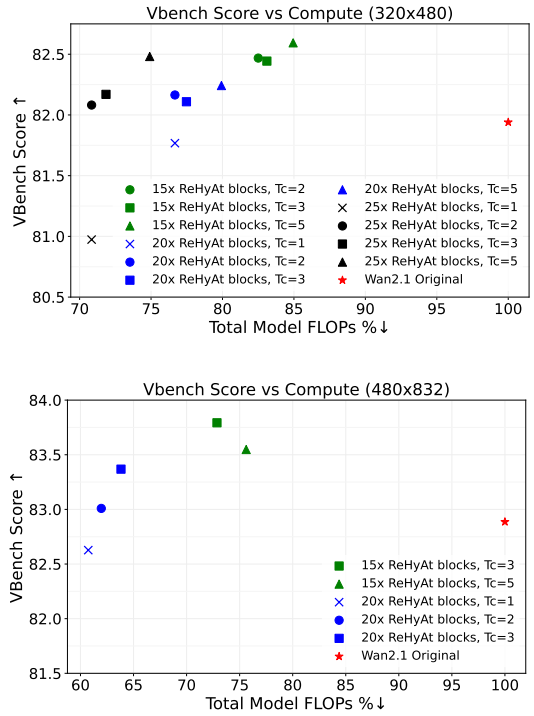


Figure 5. The total DiT FLOPs percentages versus VBench score of original Wan2.1 1.3B model compared to various hybrid configurations or 320×480 (top) and 480×832 (bottom) resolutions.

Chunk-size T_c	Block TFLOPs↓	VBench		
		Total ↑	Quality ↑	Semantic ↑
1	3.87	80.97	82.37	75.39
2	4.04	82.08	83.86	74.99
3	4.30	82.17	83.72	**75.96**
5	4.82	**82.48**	**84.12**	75.93

Table 6. Impact of T_c on ReHyAt hybrid model quality. All the models have 25× converted ReHyAt blocks with T_o=1.

model quality; however, the total score appears to saturate after that. The mild gradual improvement is still noticeable in the subject consistency dimension. This underlies the importance of overlap mechanism in decreasing temporal incoherencies.

Chunk-overlap T_o	VBench			
	Total ↑	Quality ↑	Semantic ↑	Subj. Cons.↑
0	81.56	83.23	74.90	90.90
1	82.17	83.72	**75.96**	92.05
2	82.17	83.84	75.50	92.13
3	**82.19**	**83.86**	75.51	**92.24**

Table 7. Impact of T_o on ReHyAt model quality as measured on VBench. All models have 25 converted ReHyAt blocks with T_c=3.

Causal	Block TFLOPs↓	VBench		
		Total ↑	Quality ↑	Semantic ↑
×	4.17	82.27	83.84	**75.99**
✓	**4.04**	**82.35**	**83.97**	75.87

Table 8. Impact of causality on ReHyAt hybrid model quality as measured on VBench on 15×T_c=3,T_o=0 configuration

Causality. Table 8 shows the compute and quality metrics for two equal hybrid attention formation, with causality being the only difference. We observe that the additional process to remove the non-causal attention dependency does not deteriorate the quality of the model, at least as measured by VBench. On the other hand, the saving in compute by just removing the forward-looking linear attention is not substantial. The major advantage of causal attention lies in enabling RNN reformulation, in turn enabling lower and constant peak memory and thus on-device generation of longer videos.

6. Conclusion and Future Work

In this paper, we introduced ReHyAt, a recurrent hybrid attention mechanism for video diffusion transformers that enables scalable, long-duration video generation with constant memory and linear compute requirements. Our lightweight distillation pipeline achieves near state-of-the-art quality with dramatically reduced training cost. While ReHyAt performs strongly overall, a small fraction of videos—especially with the most efficient variants—still show some temporal incoherence, highlighting an area for future improvement.

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